Universidade Federal do Amazonas
Faculdade de Engenharia Elétrica e da Computação

Atividades - Disciplina Otimização

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Março
2026

1 Tópico 1

1.1

A small engineering consulting firm has 3 senior designers available to work on the firm's 4 current projects over the next 2 weeks. Each designer has 80 hours to split among the projects, and the following table shows the manager's scoring (0 = nil to 100 = perfect) of the capability of each designer to contribute to each project, along with his estimate of the hours that each project will require.

	Project			
Designer	1	2	3	4
1	90	80	10	50
2	60	70	50	65
3	70	40	80	85
Required	70	50	85	35

(a) Formulate an allocation LP to choose an optimal work assignment.

(b) Solve your model using JuMP.

Allocation LP Formulation

Let:

$$i \in \{1, 2, 3\} \quad \text{be the designers,}$$
$$j \in \{1, 2, 3, 4\} \quad \text{be the projects.}$$

Define the decision variables:

$$x_{ij} \geq 0 \quad = \text{number of hours designer } i \text{ allocates to project } j.$$

The capability scores are given by:

$$S = \begin{bmatrix} 90 & 80 & 10 & 50 \\ 60 & 70 & 50 & 65 \\ 70 & 40 & 80 & 85 \end{bmatrix}$$

The required hours for each project are:

$$r = [70,\ 50,\ 85,\ 35]$$

Each designer has at most 80 hours available.
The LP model is:

Objective Function

Maximize the total contribution score:

$$\max Z = 90x_{11}+80x_{12}+10x_{13}+50x_{14}+60x_{21}+70x_{22}+50x_{23}+65x_{24}+70x_{31}+40x_{32}+80x_{33}+85x_{34}$$

Constraints

Designer availability constraints Each designer can work at most 80 hours:

$$\begin{aligned} x_{11} + x_{12} + x_{13} + x_{14} &\leq 80 \\ x_{21} + x_{22} + x_{23} + x_{24} &\leq 80 \\ x_{31} + x_{32} + x_{33} + x_{34} &\leq 80 \end{aligned}$$

Project requirement constraints Each project must receive exactly its required number of hours:

$$\begin{aligned} x_{11} + x_{21} + x_{31} &= 70 \\ x_{12} + x_{22} + x_{32} &= 50 \\ x_{13} + x_{23} + x_{33} &= 85 \\ x_{14} + x_{24} + x_{34} &= 35 \end{aligned}$$

Nonnegativity constraints

$$x_{ij} \geq 0, \qquad \forall i = 1, 2, 3,\ \forall j = 1, 2, 3, 4$$

Compact Form

The model can also be written compactly as:

$$\begin{aligned} \max \quad & \sum_{i=1}^{3} \sum_{j=1}^{4} s_{ij} x_{ij} \\ \text{s.t.} \quad & \sum_{j=1}^{4} x_{ij} \leq 80, \qquad \forall i = 1, 2, 3 \\ & \sum_{i=1}^{3} x_{ij} = r_j, \qquad \forall j = 1, 2, 3, 4 \\ & x_{ij} \geq 0, \qquad \forall i = 1, 2, 3,\ \forall j = 1, 2, 3, 4 \end{aligned}$$

where s_{ij} is the capability score of designer i on project j, and r_j is the number of required hours for project j.

1.2

Prove the following results. Hint: Use the definition of convex functions, the first and second order characterizations.

(a) Univariate functions:

- Exponential function: e^{ax} is convex for any a over $\mathbb{R}$
- Power function: x^a is convex for $a \geq 1$ or $a \leq 0$ over $\mathbb{R}_+$ (nonnegative reals)
- Power function: x^a is concave for $0 \leq a \leq 1$ over $\mathbb{R}_+$
- Logarithmic function: $\log x$ is concave over $\mathbb{R}_{++}$

(b) Affine function: $a^T x + b$ is both convex and concave

(c) Quadratic function: $\frac{1}{2}x^T Q x + b^T x + c$ is convex provided that $Q \succeq 0$ (positive semidefinite)

(d) Least squares loss: $\|y - Ax\|_2^2$ is always convex (since $A^T A$ is always positive semidefinite)